

The background of the slide is a dramatic, high-contrast image of intense fire. Bright orange and yellow flames are visible on the left and right sides, with a dark, smoky center. Numerous small, glowing sparks or embers are scattered throughout the dark area, creating a sense of heat and danger.

Fire-Guard Robot Saving Firefighters' Lives

"Better to melt a robot than burn a firefighter."

TEAM 9



How We Reduce Uncertainty

"Never send a firefighter into a situation where a robot could go first."



Vision Through Smoke

Thermal imaging reveals heat signatures in low/zero visibility, raising victim likelihood and identifying hot zones.



Mapping When Possible

LiDAR builds a real-time map when conditions allow, reducing uncertainty about routes, obstacles, and entry/exit paths. When mapping degrades, Fire-Guard switches to Safe Mode.



Life-Saving Data

All critical information visualized on the Commander's tablet: victim likelihood, hazard zones, last update time, and routes when available.



The Cost of Uncertainty

Worcester Cold Storage Fire, 1999

When: June 17, 1999

Where: Worcester, Massachusetts

Casualties: 6 firefighters killed

Cause: Blind entry into unknown structure. Floor collapsed under them.

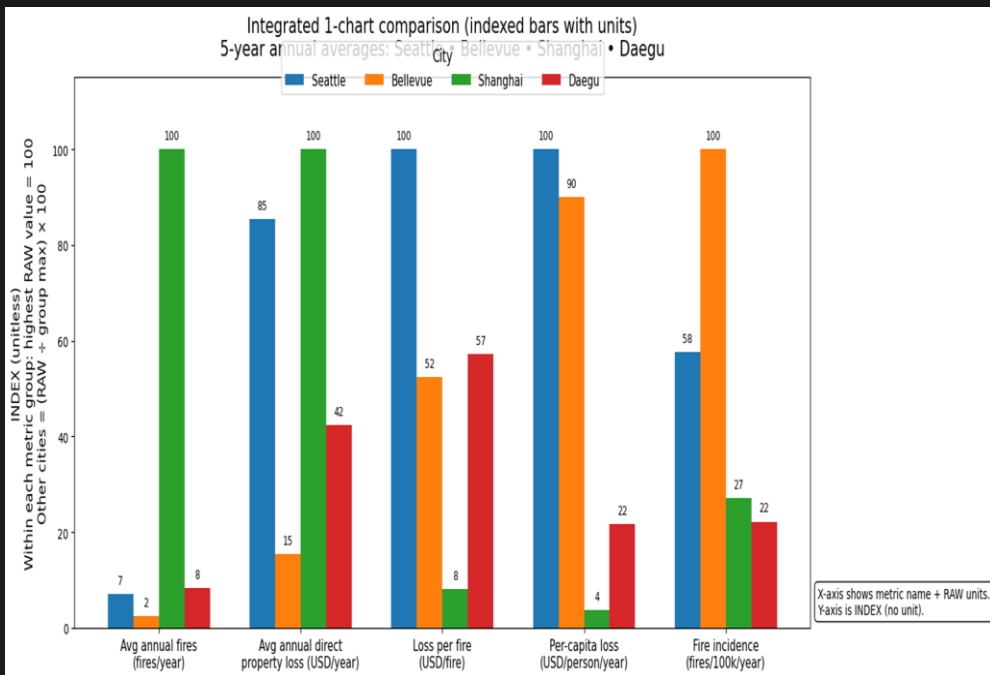
"Six firefighters entered without knowing the building's structural integrity. No one could see through the smoke. No one knew where the floor was safe."

We don't exploit this tragedy—we learn from it. Our mission is to prevent the next Worcester by giving firefighters the information they need before entry.

This is why we exist.

Fire Risk is Local & Global

Comparative Fire Statistics: Bellevue vs. Global Cities



1. Highest Fire Risk

2. High Financial Impact

3. Severe Event Intensity

4. The Solution: Speed

• Conclusion

- We face **Frequent Fires & High Costs**.
- **Early detection** is not an option, it is a necessity for our city.

We cannot change the statistics overnight,
But Fire-Guard can change the outcome.

Decisions Made in the Dark

The Commander's Dilemma



Zero Visibility

Cannot see through thick smoke.



Uncertain Victim Location

Relying on "maybe" and "might be."



Unknown Structural Risks

Floor collapses and fire spread paths unknown.



Time Pressure

Urgent decisions vs. lack of critical data.

Firefighters risk their lives entering uncertainty for "potential survivors." This is heroic, but often preventable.

Our Mission

Fill the information gap so firefighters can operate safely based on data.

Why Current Firefighting Robots Fail

The Paradox of Expensive Robots

The Problem

Many existing robots cost \$40,000–\$150,000+, making departments hesitant to deploy them in high-risk scenarios. They're too valuable to risk. So they stay in the truck.

High-Cost Ground Robot (Competitor)

Price: \$150,000+

Weight: 68 kg (150 lbs)

Issue: Too expensive to risk. Limited deployment.

Thermal Drone (Competitor)

Price: \$40,000–\$80,000

Weight: 2–5 kg

Issue: Cannot operate indoors due to smoke interference and GPS loss.

FireGuard

Price: \$3,000 per unit

Weight: 3–5 kg

Advantage: Affordable to deploy. Designed to be expendable.

FireGuard breaks the paradox: By making robots cheap enough to deploy without hesitation, we eliminate the barrier to deployment. Firefighters get the information they need, when they need it.



The Firefighter's Eyes: Seeing the Unseen

"We built a robot designed to be expendable—and we are proud of it."

Design Philosophy

- 🔥 **Low Cost:** \$3,000 per unit enables risk-free deployment.
- 🔥 **Lightweight:** 3–5 kg for rapid deployment.
- 🔥 **High-Risk Scout:** First one in, so firefighters don't have to be.

Safety-by-Design Features

- Protected core interfaces (no exposed USB ports)
- Sacrificial harness + tethered retrieval for quick swap
- Fail-safe shutdown to prevent uncontrolled operation

The Value Proposition

- Robot destroyed = \$3,000 loss
- Firefighter enters = Life at risk

The choice is clear.

"Better to melt 10 robots than burn one firefighter."



Situational Awareness Tool



Vision Through Smoke

Thermal imaging reveals heat signatures in low/zero visibility, raising victim likelihood and identifying hot zones in real time.



Mapping the Maze

LiDAR maps routes and obstacles when conditions allow; when mapping degrades, Fire-Guard switches to Safe Mode.



Life-Saving Data

All data visualized on the Commander's tablet: victim likelihood, hazard zones, last update time, and routes when available.

Firefighters check the map before they enter.



Decision Support for Incident Commanders

Data-Driven Certainty

The IC's burden: Ordering "Entry" with incomplete information.



Is there a victim inside?

Increase confidence from "unknown" to "likely/confirmed" via thermal detection.



Is the floor stable?

Visual structural assessment reduces uncertainty about safe entry zones.



Where is the fire core?

Thermal data pinpoints hottest zones to guide tactical decisions.



Decision support tool (not sole source of truth). IC retains final authority.

"The robot sees what we can't. We command with certainty, not hope."

When the Robot Finds the Path, Firefighters Don't Have to Guess

Real-world Operation Scenario



1. Deployment

Robot enters the zero-visibility smoke zone first.



2. Detection

Thermal camera identifies victim's heat signature.



3. Mapping

LiDAR builds a real-time map of the structure.



4. Report

Critical data transmitted to Commander's tablet.

*"This robot is finding the path so a firefighter **doesn't have to guess.**"*

Swarm Strategy

Power in Numbers (Staged Deployment)

\$50,000 Competitor (1 Unit) 1x Coverage	VS	\$50,000 FireGuard (5-Unit Starter Kit) 5x Coverage Expansion to 10–15 units available
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 **Multi-Angle Recon**

Deploy 5 units simultaneously to scout from different angles. Comprehensive coverage of complex structures without hesitation.

 **Risk Distribution**

Mission continues even if one unit is lost. Redundancy ensures operational success and mission resilience in harsh conditions.

 **Staged Expansion**

Start with a 5-unit starter kit per station. Scale based on operational feedback and replacement rates.

Practical Today, Perfect Tomorrow

Ready to Deploy vs. Continuous Improvement

NOW

Ready to Deploy



Price Competitiveness

Mass-deployable low-cost structure (\$3,000 per unit).



Immediate Readiness

Operational prototype with proven sensors.

Prototype v0.2: Thermal streaming live, LiDAR SLAM demo working, teleop control tested

Single-operator teleop (training < 30 min)



Proven Sensors

Reliable Thermal ($\pm 2^{\circ}\text{C}$) + LiDAR (5cm accuracy) combination.

Field-tested in simulated fire environments

FUTURE

Roadmap



Fire-Retardant Skin

Enhanced durability in extreme heat (target: 500°C / 5 min).



AI Autonomous Nav

Smarter pathfinding for complex structures. Reduced operator workload.



Deep Comms

Enhanced data transmission in shielded/underground environments (target: 200m range).

Our Design Philosophy

"We designed it to be cheap enough to melt. If it finds a victim and then melts, it has fulfilled its purpose 100%. That is a success, not a failure."

\$3,000 vs One Life

Not only a technology problem—a deployment and priority problem.

Which do you choose: A melted robot or a
burned firefighter?

Which is more important: A **\$3,000 loss** or a
human life?

Which do you want: **Heroic sacrifice** in
uncertainty or **safe rescue** based on data?

"Let's not repeat the tragedies of the past.

Join us in a new era where robots go first,
and firefighters only enter when it's certain."

Go-to-Market

Pilot-Ready Deployment & B2G Adoption Path

Customer Segments

Primary: Fire departments (100+ units in US)

Secondary: Training academies, USAR teams

Buyer: Fire Chief / Procurement Officer

Champion: Incident Commander / Training Officer

Budget: Equipment/training/safety innovation grants

Sales Motion

Phase 1 (Months 1–3): Pilot deployment with 1 fire department

Phase 2 (Months 4–6): Station package rollout (5–10 units)

Phase 3 (Months 7–12): Citywide/regional expansion

Proof Point: Measurable KPI improvements → budget approval

Pilot KPIs

- **Time-to-first-map:** <2 min
- **Victim detection:** 80%+ confidence
- **IC decision time:** 30% reduction
- **Near-miss reduction:** Measurable decrease
- **Usability:** >8/10 score

Pilot Success = Budget Approval

If pilot KPIs are met → Fire Chief approves station package purchase → Recurring revenue from replacements + software dashboard subscription

Realistic Timeline

- Pilot: 3 months (1 fire department, 5 units)
- Station rollout: 6 months (10–15 fire departments, 50–75 units)
- Year 2: Regional expansion (3–5 states, 500+ units)



Thank you!!